

drcmd: **Doubly-Robust Causal Inference with Missing Data**

Harvard Biostatistics Student Seminar Series

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Background

Unifying framework

Using the `drcmd` package

Discussion

Major goal of causal inference

Causal inference devotes large degree of attention to estimation of **counterfactual means**
 $\mathbb{E}[Y(a)]$

- If I **intervened** on a population and **gave everyone treatment value $A = a$** , what would the average outcome Y be?

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Given estimates, focus typically turns to contrasts, e.g.

$$\hat{\mathbb{E}}[Y(1)] - \hat{\mathbb{E}}[Y(0)] \quad (\text{The average treatment effect})$$

$$\hat{\mathbb{E}}[Y(1)] / \hat{\mathbb{E}}[Y(0)] \quad (\text{The causal risk ratio if } Y \text{ is binary})$$

$$\hat{\mathbb{E}}[Y(1) - Y(0) | A = 1] \quad (\text{The average treatment effect among the treated})$$

Many useful software packages already exist!

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$$\mathbb{E}[Y(a)] = \psi_a = \mathbb{E}[\mathbb{E}(Y|A = a, \mathbf{X})] \quad (\text{via causal assumptions})$$

$$\hat{\psi}_a = \frac{1}{n} \sum_{i=1}^n \left\{ \hat{\mathbb{E}}(Y_i | A_i = a, \mathbf{X}_i) + \frac{A_i}{\hat{\mathbb{P}}(A_i = a | \mathbf{X}_i)} (Y_i - \hat{\mathbb{E}}(Y_i | A_i = a, \mathbf{X}_i)) \right\}$$

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Above estimator is often called an augmented inverse probability weighted (AIPW) estimator

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- Can fit nuisance models with ML methods and still construct **valid confidence intervals**
- Provides degree of protection against model mis-specification
- Tons of software out there for implementing above estimator + its variants

Common challenge: missing data

In practice, data are often subject to **missingness**

- Data not collected for study purposes (e.g. EHR)
- Budget (certain variables often too expensive to measure for all subjects)
- Non-response, censoring

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Can categorize missingness into 3 categories: outcome, treatment, and covariates

Example: missing outcome

Y	A	X	R
NA	A_1	X_1	0
Y_2	A_2	X_2	1
Y_3	A_3	X_3	1
NA	A_4	X_4	0
Y_5	A_5	X_5	1

Example: missing treatment and outcome

Y	A	X	R
NA	NA	X_1	0
Y_2	A_2	X_2	1
Y_3	A_3	X_3	1
NA	NA	X_4	0
Y_5	A_5	X_5	1

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NA	A_4	X_4	0
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Common problem, many instance-specific solutions

At their core, these problems are *closely related*

- We have an idealized observational dataset $(Y, A, \mathbf{X})$ that is subject to missingness
- Scenarios only differ in *what* is missing, *how*, and if any other variables are available

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Construction of “doubly-robust” estimators

It turns out that by following standard approaches + theory, slight tweaks to the missingness problem can result in **significantly different** estimators

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causal estimand
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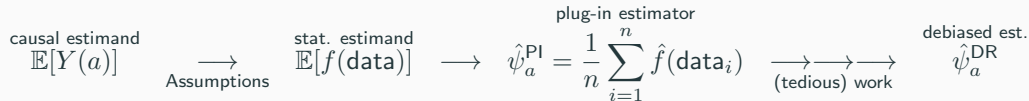
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Y_5	A_5	$\mathbf{X}_5$	1

If the MAR assumption $Y \perp\!\!\!\perp R | \mathbf{X}, A$ holds, then $\mathbb{E}[Y(a)] = \mathbb{E}[\mathbb{E}(Y | A = a, R = 1, \mathbf{X})]$ and

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is a valid estimator which inherits numerous “double-robustness” properties

Example: missing outcome with surrogates

Y	S	A	X	R
NA	S_1	A_1	X_1	0
Y_2	S_2	A_2	X_2	1
Y_3	S_3	A_3	X_3	1
NA	S_4	A_4	X_4	0
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If $Y \perp\!\!\!\perp R | \mathbf{X}, A, S$, then $\mathbb{E}[Y(a)] = \mathbb{E}[\mathbb{E}[\mathbb{E}(Y | A = a, \mathbf{X}, S, R = 1) | \mathbf{X}]]$ and

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is a valid estimator which inherits numerous “double-robustness” properties

- Nearly the same missing data structure as previous slide, yet estimator is **substantially different**

Common problem, many instance-specific solutions

Small tweaks to the missing data problem can lead to significant changes to the corresponding statistical estimand / estimator

Frustrating issue, since it *decentralizes* resources applied researchers can turn to

- No single R package, and often no packages available *at all*, for specific applications
- Multiple imputation (`mice`) is general, but lacks theoretical guarantees often desired of DR-style estimators

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It turns out there is a unifying framework which **links** different missing data problems

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High-level idea: Connect the estimator we'd **ideally** construct, under no missing data, to the observed data distribution

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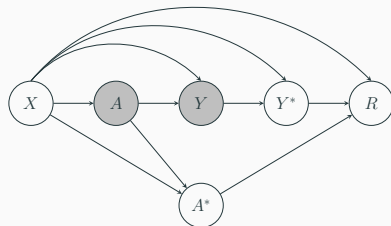
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High-level idea: Connect the estimator we'd **ideally** construct, under no missing data, to the observed data distribution

- Underlying theory was extensively developed by Andrea Rotnitzky, Jamie Robins, Mark van der Laan and others in the 90s + early 2000s, but remains relatively underutilized

Unifying framework example: missing outcome + treatment

Y	Y^*	A	A^*	X	R
Y_1	Y_1^*	A_1	A_1^*	X_1	1
	Y_2^*		A_2^*	X_2	0
	Y_3^*		A_3^*	X_3	0
Y_4	Y_4^*	A_4	A_4^*	X_4	1
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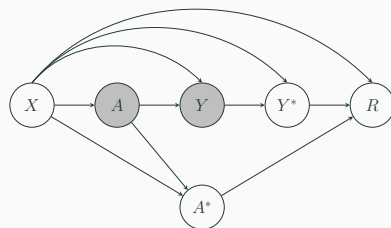
Example problem: Suppose we observe

$$\mathbf{O}_i = (Y_i, Y_i^*, A_i, A_i^*, \mathbf{X}_i, R_i)$$

and want to estimate $\mathbb{E}[Y(a)]$

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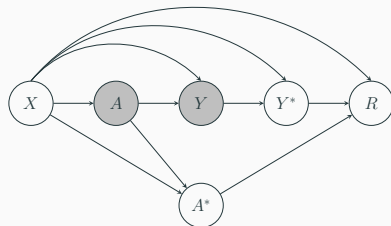
$$\mathbf{O}_i = (Y_i, Y_i^*, A_i, A_i^*, \mathbf{X}_i, R_i)$$

and want to estimate $\mathbb{E}[Y(a)]$

- Y_i and A_i are **expensive** to measure, and only observed when $R_i = 1$
- Inexpensive, error-prone measurements of Y_i and A_i , denoted Y_i^* and A_i^* , are available for all subjects

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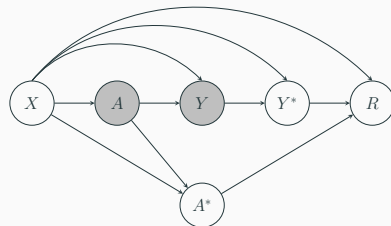
Idea: If we had complete data, could construct a plug-in estimator $\hat{\psi}_a^{\text{PI}} = \frac{1}{n} \sum_{i=1}^n \hat{m}_a(\mathbf{X}_i)$,

$$\hat{\psi}_a^{\text{PI}} = \hat{\psi}_a^{\text{PI}} = \frac{1}{n} \sum_{i=1}^n \left(m_a(\mathbf{X}_i) + \frac{I(A_i = a)}{p_a(\mathbf{X}_i)} (Y_i - m_a(\mathbf{X}_i)) - \psi_a^{\text{PI}} \right)$$

where $m_a(\mathbf{X}) = \mathbb{E}(Y|A = a, \mathbf{X})$ and $p_a(\mathbf{X}) = \mathbb{P}(A = a|\mathbf{X})$

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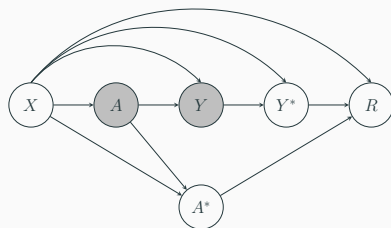
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$$\hat{\psi}_a^{\text{DR}} = \hat{\psi}_a^{\text{PI}} + \frac{1}{n} \sum_{i=1}^n \left(\hat{m}_a(\mathbf{X}_i) + \frac{I(A_i = a)}{\hat{g}_a(\mathbf{X}_i)} \{Y_i - \hat{m}_a(\mathbf{X}_i)\} - \hat{\psi}_a^{\text{PI}} \right)$$

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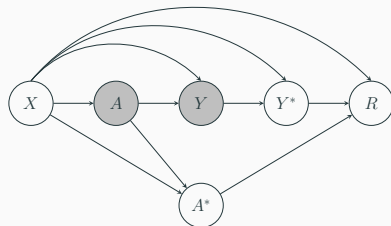
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Above, $\hat{m}_a(\mathbf{X})$ and $\hat{g}_a(\mathbf{X})$ can be estimated with weighted regressions that add weights $R/\hat{\mathbb{P}}(R=1|\mathbf{Z})$ to the underlying loss function

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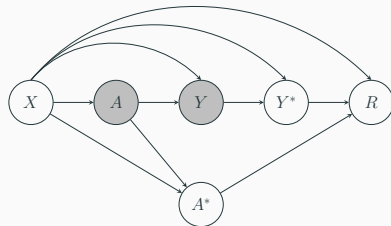
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Issue: The bias correction terms are only observed when $R_i = 1$

- Above estimator is **infeasible**

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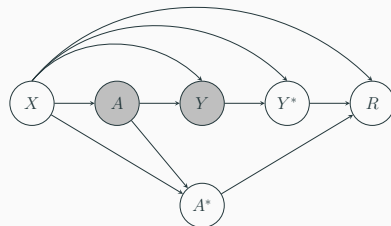


Key idea: Treat the bias correction terms as **pseudo-outcomes**:

$$\begin{aligned}\hat{\psi}_a^{\text{DR}} &= \hat{\psi}_a^{\text{PI}} + \frac{1}{n} \sum_{i=1}^n \left(\hat{m}_a(\mathbf{X}_i) + \frac{I(A_i = a)}{\hat{g}_a(\mathbf{X}_i)} \{Y_i - \hat{m}_a(\mathbf{X}_i)\} - \hat{\psi}_a^{\text{PI}} \right) \\ &= \hat{\psi}_a^{\text{PI}} + \frac{1}{n} \sum_{i=1}^n \underbrace{\chi_a(\mathbf{O}_i; \hat{m}_a, \hat{g}_a)}_{\text{pseudo outcome}}\end{aligned}$$

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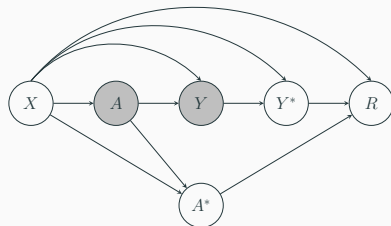


Key idea: Perform AIPW on the **pseudo-outcome**:

$$\hat{\psi}_a^{\text{DR}} = \hat{\psi}_a^{\text{PI}} + \frac{1}{n} \sum_{i=1}^n \chi_a(\mathbf{O}_i; \hat{m}_a, \hat{g}_a)$$

Unifying framework example: missing outcome + treatment

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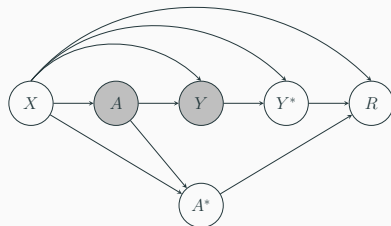


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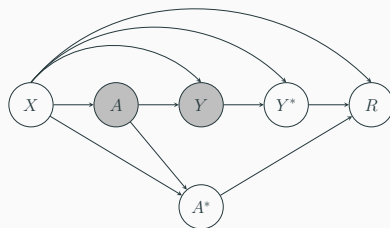
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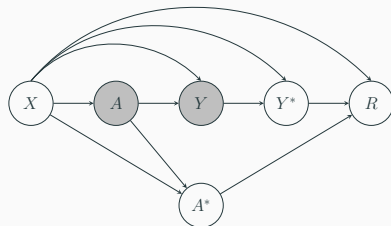
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- Feasible estimator!
- When sampling probabilities are known, $\hat{\psi}_a^{\text{DR}}$ is doubly-robust in the traditional sense

Unifying framework example: missing outcome + treatment

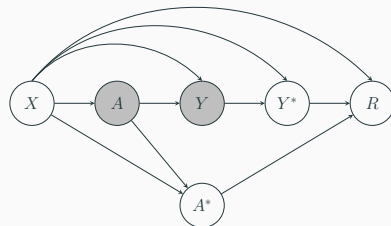
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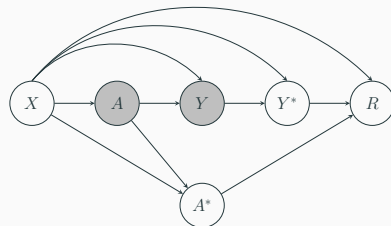


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1. Fit m_a and g_a on the subset of data with complete cases via weighted regressions, allowing us to form pseudo-outcomes χ_a

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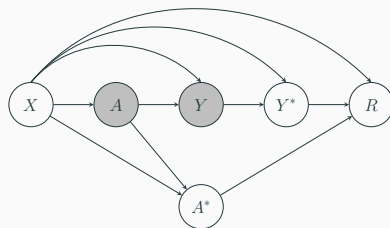


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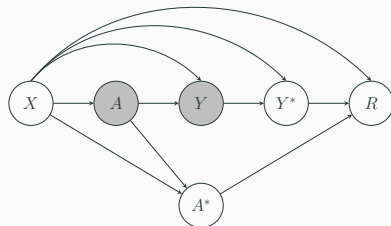


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3. Perform AIPW on the pseudo-outcomes

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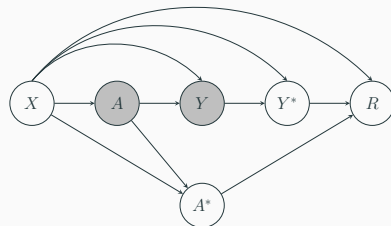
The missing data structure will determine what variables Z are never missing

- But approach can be applied to **more general** missing data problems
- Requires

Partially missing variables $\perp\!\!\!\perp R|Z$

Unifying framework example: missing outcome + treatment

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The missing data structure will determine what variables $\mathbf{Z}$ are never missing

- But approach can be applied to more general missing data problems
- Requires

$$(Y, A) \perp\!\!\!\perp R \mid (\mathbf{X}, A^*, Y^*)$$

Background

Unifying framework

Using the `drcmd` package

Discussion

The drcmd function

```
drcmd <- function(Y,A,X,  
                  W=NULL,  
                  learners='SL.glm',  
                  eem_ind=FALSE,  
                  k=1,  
                  ...) {  
  
  <automatically determine missingness pattern>  
  
  <use pattern + DR-theory to obtain DR estimators>  
  
  <form the results structure>  
  
  return(results)  
}
```

Getting started with drcmd

Calling drcmd is relatively **simple**:

```
results <- drcmd::drcmd(Y,A,X, default_learners='SL.glm')
```

And can be **customized**:

```
results <- drcmd::drcmd(Y,A,X,W=data.frame(Ystar),  
                        learners=c('SL.xgboost','SL.glm',  
                                   'SL.ranger'),  
                        k=10, tml=TRUE)
```

User has control over means by which nuisance functions are fit, whether to obtain analytical or bootstrap SEs, whether to perform cross-fitting, and additional tweaks to the estimation procedure

Working with drcmd output

drcmd outputs an R S3 object which stores results

- Allows the use of base functions like `print`, `summary`, etc. to directly read in drcmd output

```
> summary(results)
```

```
=====
                        Summary of drcmd results
=====
```

Metric	Estimate	SE
ATE:	0.890	0.117
E[Y(1)]:	0.978	0.097
E[Y(0)]:	0.088	0.079

```
-----
Variables with missingness (U): Y
Variables without missingness (Z): X, A
-----

Validity of results requires causal assumptions to hold,
as well as the assumption that U is independent of
R given Z
```

Comparison with other packages

While existing DR causal packages don't support *general* missing data, a few support **missing outcomes** under the assumption $Y \perp\!\!\!\perp R | X, A$

- AIPW
- tmle
- drtmle

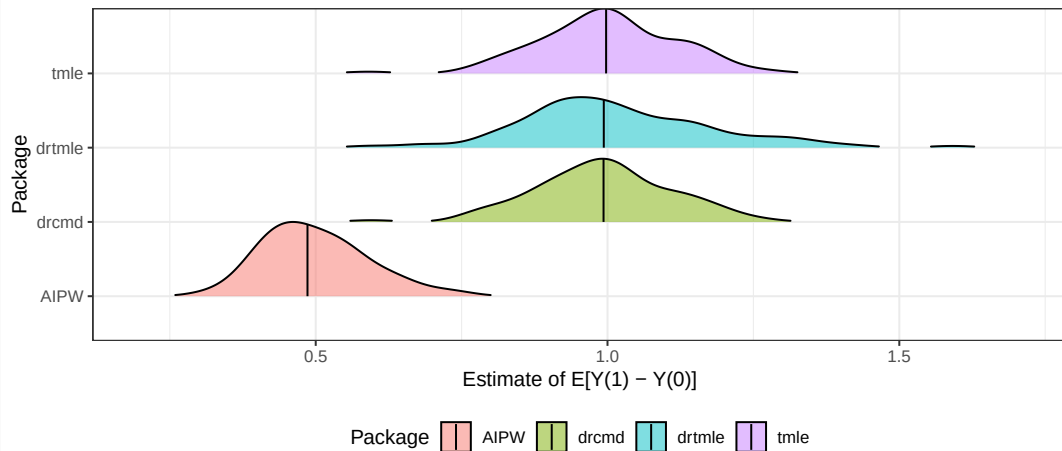
Compared drcmd to these packages under a simple outcome MAR setting

- Used same stacked learners to fit all nuisance functions
- Interested in relative statistical + computational efficiency

Results: estimation of ATE

Comparison of DR causal inference packages: estimation

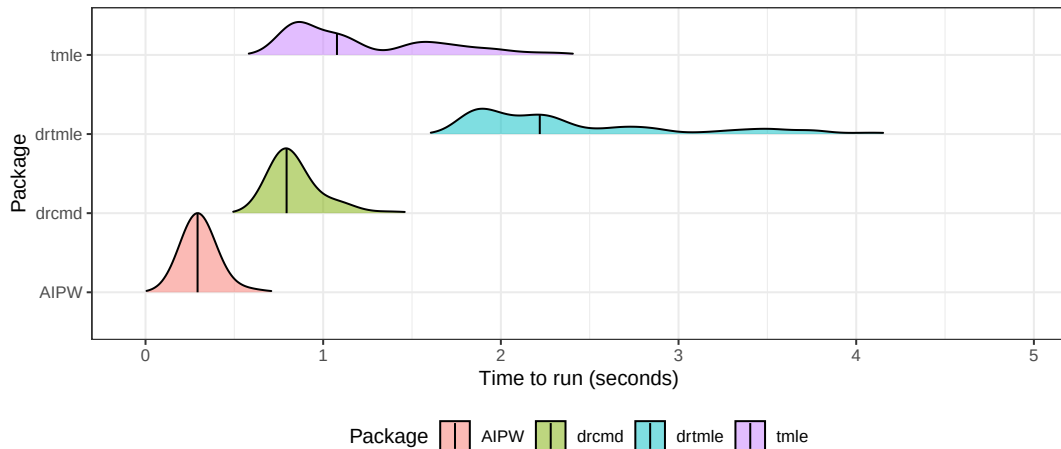
Outcome missing at random conditional on measured covariates



Results: speed

Comparison of DR causal inference packages: time to run

Outcome missing at random conditional on measured covariates



Background

Unifying framework

Using the `drcmd` package

Discussion

Many existing packages for performing doubly-robust causal inference in R, but...

- Ability to handle missing data is **severely limited**, and...
- Missing data is typically **the rule**, rather than the exception

drcmd functions to fill this gap...

- Able to determine missingness structure + construct appropriate DR estimators **on the fly**
- Simulation studies for missing outcome setting suggest drcmd performs competitively with existing packages in estimation efficiency and runtime
 - With added benefit of being able to handle **far more complex** missing data structures

Appendix

Unifying framework

It turns out there is a unifying framework which links different missing data problems

Fixate interest on estimating $\mathbb{E}[Y(a)]$, briefly assume we observe $O_i = (Y_i, A_i, \mathbf{X}_i)_{i=1}^n$

Unifying framework

It turns out there is a unifying framework which links different missing data problems

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Recall the doubly-robust estimator

$$\hat{\psi}_a = \frac{1}{n} \sum_{i=1}^n \left\{ \hat{\mathbb{E}}(Y_i | A_i = a, \mathbf{X}_i) + \frac{A_i}{\hat{\mathbb{P}}(A_i = a | \mathbf{X}_i)} (Y_i - \hat{\mathbb{E}}(Y_i | A_i = a, \mathbf{X}_i)) \right\}$$

Amounts to fitting an outcome regression, and adding on the IP-weighted residuals from that regression

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$$\hat{\psi}_a = \frac{1}{n} \sum_{i=1}^n \left\{ \hat{m}_a(\mathbf{X}_i) + \frac{A_i}{\hat{g}_a(\mathbf{X}_i)} (Y_i - \hat{m}_a(\mathbf{X}_i)) \right\}$$

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Recall the doubly-robust estimator of $\psi_a = \mathbb{E}[\mathbb{E}(Y|A = a, \mathbf{X})]$

$$\hat{\psi}_a = \frac{1}{n} \sum_{i=1}^n \chi_a^F(O_i; \hat{m}_a, \hat{g}_a)$$

$\chi_a^F(O_i; \hat{m}_a, \hat{g}_a)$ is referred to as the efficient influence function (EIF) for estimating ψ_a

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EIFs are **crucial ingredients** for constructing DR estimators

- Different statistical estimands have different EIFs

Major result for missing data

Suppose interest lies in estimating $\psi_a = \mathbb{E}[\mathbb{E}(Y|A = a, \mathbf{X})]$. Let $\chi_a^F(O; m_a, g_a)$ denote the EIF of this estimand we'd target under full (missingness-free) data

Let the **ideal** data O be split into always-observed and sometimes-missing components: $O = (V, U)$, and suppose in practice we actually observe

$$(W, \quad V, \quad UR, \quad R)$$

where U is only available if $R = 1$. Let $\mathbf{Z} = (W, V)$ denote variables always observed by the analyst

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If $U \perp\!\!\!\perp R | \mathbf{Z}$, then it turns out the EIF for ψ_a with our observed data can be written

$$\mathbb{E}[\chi_a^F(O; m_a, g_a) | \mathbf{Z}] + \frac{R}{\hat{\mathbb{P}}(R = 1 | \mathbf{Z})} (\chi_a^F(O; m_a, g_a) - \mathbb{E}[\chi_a^F(O; m_a, g_a) | \mathbf{Z}])$$

Unifying framework

Why is this useful? Implies a straightforward + **general** procedure for estimation of causal effects (given plausibility of the MAR assumption)

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- Final estimate looks like AIPW, where the $\chi_a^F(O; \hat{m}_a, \hat{g}_a)$ are “pseudo-outcomes”

$$\frac{1}{n} \sum_{i=1}^n \left(\hat{\mathbb{E}}[\chi_a^F(O_i; \hat{m}_a, \hat{g}_a)|\mathbf{Z}_i] + \frac{R_i}{\hat{\mathbb{P}}(R_i = 1|\mathbf{Z}_i)} (\chi_a^F(O_i; \hat{m}_a, \hat{g}_a) - \hat{\mathbb{E}}[\chi_a^F(O_i; \hat{m}_a, \hat{g}_a)|\mathbf{Z}_i]) \right)$$

Example: missing outcome with surrogates

Return to the [missing outcome](#) example where we observe

$$(R_i Y_i, A_i, \mathbf{X}_i, S_i, R_i), \quad i = 1, \dots, n$$

Here, our [ideal](#) data structure would be $O = (Y, A, X)$. Recal the observed data structure formulation as (W, V, UR, R) , and notice

$$W = S, \quad V = (\mathbf{X}, A), \quad U = Y, \quad \text{and} \quad \mathbf{Z} = (\mathbf{X}, A, S)$$

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1. Regress R on $\mathbf{Z}$ to obtain an estimate of $\hat{\mathbb{P}}(R = 1|\mathbf{Z})$

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2. Estimate $\hat{m}_a$ and $\hat{g}_a$ among the complete cases, adding weights $R_i / \hat{\mathbb{P}}(R_i = 1 | Z_i)$ to corresponding loss functions

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$$W = S, \quad V = (\mathbf{X}, A), \quad U = Y, \quad \text{and} \quad \mathbf{Z} = (\mathbf{X}, A, S)$$

3. Use the estimated $\hat{m}_a$ and $\hat{g}_a$ to construct pseudo-outcomes $\chi_a^F(O_i; \hat{m}_a, \hat{g}_a)$ for the complete cases

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4. Regress those pseudo-outcomes on $\mathbf{Z}$ among the complete cases, yielding an estimate of $\hat{\mathbb{E}}[\chi_a^F(O_i; \hat{m}_a, \hat{g}_a) \mid \mathbf{Z}, R = 1]$

Example: missing outcome with surrogates

Return to the [missing outcome](#) example where we observe

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$$W = S, \quad V = (\mathbf{X}, A), \quad U = Y, \quad \text{and} \quad \mathbf{Z} = (\mathbf{X}, A, S)$$

5. Form the final estimator

$$\hat{\psi}_a = \frac{1}{n} \sum_{i=1}^n \left(\mathbb{E}[\chi_a^F(O_i; \hat{m}_a, \hat{g}_a) \mid \mathbf{Z}] + \frac{R_i}{\hat{\mathbb{P}}(R_i = 1 \mid \mathbf{Z}_i)} \{ \chi_a^F(O_i; \hat{m}_a, \hat{g}_a) - \mathbb{E}[\chi_a^F(O_i; \hat{m}_a, \hat{g}_a) \mid \mathbf{Z}] \} \right)$$

Consistency conditions

Define

$$m_a(\mathbf{X}) = \mathbb{E}[Y|A = a, \mathbf{X}] \quad g_a(\mathbf{X}) = \mathbb{P}(A = a|\mathbf{X}) \quad \kappa(\mathbf{Z}) = \mathbb{P}(R = 1|\mathbf{Z})$$

$$\chi_a^F(O; m_a, g_a) = m_a(\mathbf{X}) + \frac{A}{g_a(\mathbf{X})}(Y - m_a(\mathbf{X}))$$

$$\varphi_a(\mathbf{Z}) = \mathbb{E}[\chi_a^F(O; m_a, g_a)|\mathbf{Z}, R = 1]$$

Then, it can be shown

$$\hat{\psi}_a - \psi_a = O_{\mathbb{P}} \left(\frac{1}{\sqrt{n}} + \|\hat{m}_a - m_a\| \cdot \|\hat{g}_a - g_a\| + \|\hat{\kappa} - \kappa\| \cdot \|\hat{\varphi}_a - \varphi_a\| \right)$$